

GPS-Free Self-Healing Drone Networks via Virtual Spring-Dampers

Federico Corò, Riccardo Fabbian, and Claudio E. Palazzi

Department of Mathematics

University of Padua

Padua, Italy

federico.coro@unipd.it, fabb.riccardo@gmail.com, cpalazzi@math.unipd.it

Abstract—Distributed collective intelligence enables groups of autonomous agents to cooperatively achieve system-level goals through local interactions and decentralized decision-making. In this context, this paper presents a self-healing protocol for drone swarms that restores connectivity to a base station after link loss. The protocol combines ACK-based reachability detection, flooding-based hop discovery, periodic neighbor broadcasts, and a virtual spring-damper formation controller. Localization is GPS-free and uses a UWB PHY-layer channel with anchor beacons, Time-of-Flight ranging, and least-squares trilateration. We evaluate the system in *ns-3* on 15 single-base scenarios that vary topology, swarm density, ranging noise, and scheduled drone failures. The protocol restores connectivity within ~ 1 s in every scenario; between the two attractive-force variants, healing latency is the same, while the weighted variant reduces mean return latency by 8% and its p_{95} tail by 21%.

Index Terms—Distributed Collective Intelligence, Drone Swarm, UWB, Trilateration, Localization, Self-Healing.

I. INTRODUCTION

Drone swarms represent a compelling instance of distributed collective intelligence, where autonomous agents cooperate through local sensing, communication, and decentralized decision-making to achieve system-level objectives. Compared to single drones, swarms provide larger coverage, redundancy, and modular task specialization, but they also require robust communication between drones and base stations, as well as coordinated formation control [1]. In decentralized swarm operations, communication links between individual drones and the base station may be disrupted by jamming, interference, or geographic occlusions. In such situations, the swarm must autonomously detect connectivity loss and collectively reorganize through a self-healing process that restores communication via a multi-hop relay chain, preventing network partitioning without centralized control.

To address this challenge, in this work in progress paper we present and discuss a self-healing protocol in which drones detect the loss of base connectivity and cooperatively reconfigure their positions through a virtual spring-damper controller. Localization is performed without GPS and relies on UWB Time-of-Flight trilateration against fixed anchors. We evaluate two controller variants (centroid and weighted) in *ns-3* on a single-base experimental batch that varies topology, swarm density, ranging noise, and scheduled drone failures. The main contributions of this work are: a UWB-based GPS-free

communication and localization stack, a self-healing protocol with return and station-keeping behaviors, and a comparative evaluation of the two controller variants under varying network and environmental conditions.

The rest of this paper is organized as follows. Section II provides an overview of the related literature. Section III presents the proposed self-healing solution. In Sections IV we discuss the experimental setup we employed and in Section V we discuss the outcome of our tests. In Section VI we propose the next steps of this work-in-progress, while in Section VII conclusions are drawn.

II. RELATED WORK

Restoring connectivity in mobile multi-agent systems has been studied across several research lines. In wireless sensor and actor networks, topology management surveys [2] and cascaded-relocation schemes [3] recover from node failures by triggering only the minimum number of nodes after a partition. Restore connectivity through mobile relays [4] has shown that deploying mobile actors to bridge disjoint segments is more efficient than complete redeployment, an insight that motivates self-healing of the relay-chain in aerial settings.

For unmanned aerial vehicles, formation control is based on classical flocking and potential-field methods. Olfati-Saber's flocking framework [5] and Khatib's artificial potential fields [6] provide the analytical basis that virtual spring-damper models [7], [8] extend to networked agents. Real flying swarms have demonstrated decentralized coordination using only local communication [9], [10], while Coppola *et al.* [11] showed that relative localization based on on-board communication is sufficient for safe MAV teaming.

GPS-free self-localization is a parallel concern. UWB ranging combined with anchor beacons supports centimeter-level positioning for aerial robots [12], [13]. Our contribution integrates these strands by combining UWB-only localization, a hop-aware spring-damper controller, and an explicit loss-of-link detector to autonomously restore single-base connectivity after drone failures.

III. SELF-HEALING PROTOCOL

The protocol enables a drone to detect loss of communication with the base station and autonomously reorganize to

(a) Initial configuration: L1's direct link is about to fail (b) After self-healing: relay chains restored

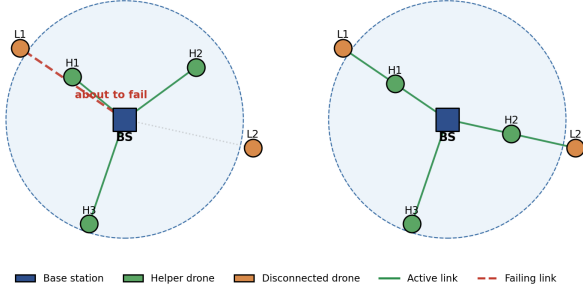


Fig. 1. Self-healing scenario. (a) Initial layout: helper drones (H1–H3) are inside coverage; L1 holds a direct link to the base that is about to fail (red dashed), while L2 is already isolated. (b) After the spring-damper repositioning, helpers move outward to bridge each disconnected drone with the base via a multi-hop relay.

form a multi-hop relay chain. It has two components: a loss-of-link detector and a formation controller based on a virtual spring-damper model. Upon ACK timeout, a drone runs the controller using neighbor positions and hop counts (obtained by flooding). The controller produces attractive forces toward the preceding and following hops in the relay chain, plus repulsive forces against close neighbors. The equilibrium is the geometric midpoint between the previous and next hop, which maximizes the Signal-to-Noise Ratio on both incident links. Figure 1 sketches a representative scenario: an in-coverage drone whose direct link is about to fail, and the post-healing relay chain that helper drones form to restore connectivity.

A. System Architecture

The architecture has three node roles: a stationary Base Station (B), static UWB Anchors (A_j), and mobile Drones (D_i) running an identical control stack. The base station also acts as an anchor. Following the Crazyflie 2.1 model [14], base traffic is unicast while swarm coordination uses broadcast. Messages are delivered over a UWB PHY-layer channel that adds propagation delay (d/c) and enforces a range cutoff. Each drone dispatches packets to four logical modules: hop-discovery flooding, a neighbor table keyed by neighbor ID, a UWB ranging/trilateration module, and core control logic for position updates, ACKs, and help-proxy requests. The controller reads UWB position and neighbor data and emits a velocity command through an actuator. The base station tracks position updates, replies with ACKs (carrying base position and hop 0), and periodically seeds floods.

B. Help-Proxy Triggering

Each drone sends periodic unicast `PositionUpdate` messages to the base and waits for an ACK. If no ACK arrives within a safety window of $1.5s$, the drone considers the link broken and broadcasts a “help-proxy” signal (`HELP_PROXY CORE` packet); any peer receiving it switches into mission mode. Broadcast matches the Crazyflie communication model.

After issuing `HELP_PROXY`, the requester is return-armed once it receives the first relayed ACK addressed to itself. Re-

turn begins after a hop-staggered timeout similar to $802.11p$, proportional to $(H_{max} - h)$ where H_{max} is max hop distance from any drone back to the base while h is the drone’s own hop count. This enables the farthest drones return first and propagate a `RETURNING` flag back along the chain, collapsing it from the tail inward. During return, the attractive gain is reduced and the drone moves toward direct coverage; on receiving a direct ACK it enters station-keeping near the coverage boundary, re-arming if coverage is lost again to prevent oscillation.

C. Flooding-Based Hop Discovery

Flooding allows drones to estimate the hop distance to the base station even when direct connectivity is lost. The base periodically broadcasts a `START`; every in-coverage drone becomes an initiator and rebroadcasts a `DISCOVERY` with hop 1. Broadcast initiation (instead of unicast to a chosen seed) avoids relying on a single node that might be out of range. On `DISCOVERY` reception, a drone with a live direct ACK declares hop 1; otherwise it increments the carried hop. A node that detects an ACK timeout suppresses its stale “hop 1” until the next flood, preventing false attraction toward a dead link. Each node only reacts when a shorter path to the base arrives for the same flood instance, then emits a `REPORT` and rebroadcasts `DISCOVERY`; reports are forwarded at most once per improvement, keeping the protocol lightweight.

D. Neighbor Information Exchange

For local, decentralized reactivity, each drone maintains a neighbor table fed by periodic `NEIGHBOR` broadcasts. The payload carries the sender ID, its hop count to the base, and UWB-trilaterated coordinates. When a drone has a direct ACK from the base, it injects a synthetic neighbor entry for the base (position from the ACK, hop 0), so the controller can treat the base as just another neighbor without special-casing.

E. Formation Control via Virtual Spring-Damper Model

Healing behavior is realized by a virtual spring-damper controller [7], [8] that combines linear attraction to neighbors at adjacent hop levels (the previous and next hops in the relay chain) with $1/d^2$ short-range repulsion against any neighbor closer than the safety distance D_{safe} :

$$\vec{F}_{tot} = \sum_{j \in N_{prev} \cup N_{next}} K_{att}(\vec{p}_j - \vec{p}_i) - \sum_{\substack{j \in N \\ d_{ij} < D_{safe}}} \frac{K_{rep}}{d_{ij}^2} \cdot \frac{\vec{p}_j - \vec{p}_i}{d_{ij}} \quad (1)$$

where $\vec{p}_i$ and $\vec{p}_j$ are the positions of drone i and neighbor j ; N_{prev} and N_{next} are the sets of neighbors at the previous and next hop levels (hop $h - 1$ and $h + 1$); N is the set of all neighbors; $d_{ij} = \|\vec{p}_i - \vec{p}_j\|$ is the Euclidean distance between drones; K_{att} and K_{rep} are the attractive and repulsive gains (Table I); and D_{safe} is the minimum safe separation distance. The attractive term drives a drone toward the midpoint of its previous and next hops; repulsion is applied to all neighbors, including peers with the same hop count. We compare two attractive variants: (i) *centroid* groups neighbors by hop count

TABLE I
BASELINE SIMULATION PARAMETER VALUES

Parameter	Value
Coverage radius R_{max}	50 m
Coverage area	Circular, centered at base station
Number of drones N_d	20
Number of anchors N_a	18
Controller algorithm	centroid, weighted
Attractive gain K_{att}	0.3
Repulsive gain K_{rep}	8.0
Safety distance D_{safe}	2.0 m
Max speed V_{max}	1.0 m/s
Drone mass m	0.029 kg
Tick interval T_{ctrl}	0.05 s
Simulation duration T_{sim}	300 s
UWB noise σ_{uwb}	0.0 m

and applies one pull per group centroid, while (ii) *weighted* scales each pull by the opposite-side population so the mid-point stays the equilibrium even under unbalanced sides. The net force is converted to acceleration via Newton’s second law and applied through a velocity actuator capped at V_{max} . A mission flag gates the controller: drones broadcast neighbor state while idle and switch on the spring-damper loop when a help-proxy trigger is received.

IV. EXPERIMENTAL SETUP

The protocol was evaluated in a custom *ns-3* [15] scenario built around a direct UWB PHY-layer channel that delivers each transmission after a d/c propagation delay with a strict range cutoff. Static UWB anchors and the base station itself periodically broadcast position-tagged beacons; drones compute Time-of-Flight ranges and estimate their pose by least-squares trilateration. The baseline scenario uses one base at (0,0,0), 20 drones, and 18 standalone anchors. Six drones act as helpers on a ~ 40 m ring just inside the 50 m coverage radius; 11 drones start in the 68–82 m belt just outside coverage; three far scouts at 112–118 m force a multi-hop chain. Anchor placement guarantees at least three line-of-sight beacons everywhere. Each drone ticks at 0.05 s, performing a position update to the base, an ACK-timeout check (with `HELP_PROXY` on miss), and a controller step. The ACK timeout is 1.5 s, position updates are sent every 0.5 s or on movement, and anchors beacon at 2 Hz. A JSON declaration file runs batches that override positions and schedule mid-simulation drone failures. Table I lists the baseline parameters.

A. Mobility Model

To focus on network-driven coordination, drones use a lightweight kinematic mobility model that integrates commanded accelerations under standard Newtonian motion:

$$\mathbf{p}_{new} = \mathbf{p}_{old} + \mathbf{v}_{old}\Delta t + \frac{1}{2}\mathbf{a}\Delta t^2, \quad \mathbf{v}_{new} = \mathbf{v}_{old} + \mathbf{a}\Delta t \quad (2)$$

where $\mathbf{p}$ and $\mathbf{v}$ are position and velocity, $\mathbf{a}$ is the acceleration computed from Eq. (2), $\Delta t = T_{ctrl} = 0.05$ s is the control tick interval, and subscripts *old* and *new* are consecutive time steps. Trajectories are exported via *ns-3*’s `AnimationInterface` for inspection in NetAnim [?].

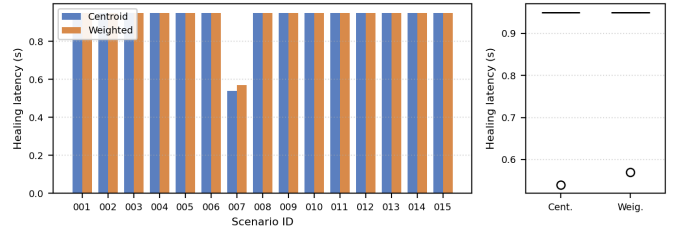


Fig. 2. Healing-latency distribution across the 15 scenarios. Left: per-scenario values for both controllers. Right: aggregate boxplot.

V. RESULTS AND ANALYSIS

We ran a single-base batch of 15 scenarios that covers baseline, dense, linear, tight, wide, and asymmetrically skewed swarm topologies; sparse anchor configurations; noisy ranging; heavy/slow drones; and scheduled drone failures (helper, relay, cascade, simultaneous-cluster kills, and a kill timed during the healing window). Each scenario was run exactly twice with deterministic configuration: once with the centroid controller and once with the weighted controller, for a total of 30 runs. All runs completed successfully. The simulator reports trilateration error, healing latency (time from `HELP_PROXY` to the first relayed ACK), return latency, recovery rate, final hop coverage, and station-keeping drift. Table II summarizes the batch.

TABLE II
BATCH METRICS ACROSS THE 15 SINGLE-BASE SCENARIOS (MEAN $\pm$ STD. DEV.).

Metric	Centroid	Weighted
Trilateration avg err. (m)	0.4708 $\pm$ 0.2526	0.4308 $\pm$ 0.2261
Healing latency avg (s)	0.9227 $\pm$ 0.1059	0.9247 $\pm$ 0.0981
Return latency avg (s)	23.2436 $\pm$ 12.04	21.36 $\pm$ 6.19
Recovery rate (%)	85.24 $\pm$ 28.11	83.84 $\pm$ 33.24
Final coverage (%)	98.63 $\pm$ 3.607	98.63 $\pm$ 3.607

The protocol restores connectivity in roughly one second in every batch. Figure 2 shows that the two variants of the controller are tied in this metric: every scenario except *noisy_uwb_long* (007) sits on the hop-staggered wait floor of 0.95 s for both algorithms, and (007) itself differs by less than 0.04 s. The wait dominates the bridging time within a single base’s coverage radius, so the centroid/weighted difference does not show up here.

The two variants do diverge on the *return* phase, where the lost drones move back into direct coverage. The weighted controller cuts mean return latency by 8% (21.4 s vs 23.2 s) and the p_{95} tail by 21% (27.6 s vs 35.1 s); its return-path efficiency is also tighter (1.05 ± 0.07 vs 1.12 ± 0.28). The recovery rate and the final coverage are similar. A simple reason is that the weighted scaling dampens the gradient as the chain collapses inward, reducing the late overshoot. The comparison under partitioned topologies is deferred to the multi-base extension (Section VI).

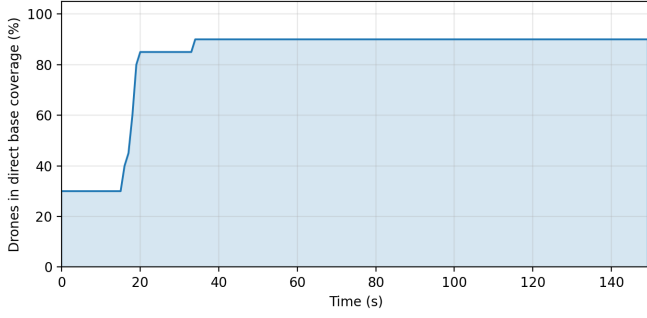


Fig. 3. Node connections over time in a representative run.

A. Parameter Sensitivity

The controller gains $K_{att} = 0.3$, $K_{rep} = 8.0$, and $D_{safe} = 2.0$ m were selected by a grid search that, for each candidate triple, scored oscillation amplitude, time-to-target, and minimum inter-drone distance. In practice, increasing K_{att} shortens convergence time, but causes relay drones to overshoot the midpoint, especially when the velocity cap V_{max} saturates and the drone cannot decelerate before crossing. Reducing K_{att} from the previous tested value of 1.0 to 0.3 removed visible oscillations at the cost of about 1 s of extra settling time. In contrast, raising K_{rep} or D_{safe} widens the safety margin, but when repulsion dominates attraction, helpers fail to compress into a relay chain near the boundary, and the lost drone is never bridged. Drone mass acts like damping through $\mathbf{a} = \mathbf{F}_{tot}/m$: heavier drones smooth the response but slow convergence. The chosen triple sits where oscillations are damped, the chain still compresses, and time-to-target stays bounded.

Figure 3 traces active connections during a representative run, showing the transition from isolated links to a restored multi-hop relay.

VI. FUTURE WORK

This is a work-in-progress and several extensions are planned for the full version. *Multi-base topologies*: extend the flooding and neighbor exchange to track per-base hop counts independently and let each drone optimize against its nearest reachable base. *Base-station failure tolerance*: add scheduled base kills to the batch and quantify recovery as drones reattach to surviving bases, including cascading scenarios where multiple bases fail in sequence. *Channel realism*: replace the strict range-cutoff UWB model with a path-loss/fading propagation model. *Parameter analysis*: report a complete sensitivity sweep over $(K_{att}, K_{rep}, D_{safe})$ rather than the qualitative summary given here, including the trade-off between healing latency and station-keeping drift. *Network scale*: extend the evaluation beyond 20 drones and include heterogeneous swarms with mixed roles. *Hardware validation*: port the stack to Crazyflie 2.1 with Loco UWB anchors and reproduce the baseline scenario in a flying arena to quantify the sim-to-real gap.

VII. CONCLUSIONS

In this work in progress paper, we presented a decentralized self-healing protocol that combines ACK-based reachability detection, flooding-based hop discovery, neighbor broadcasts, and a virtual spring-damper controller, with GPS-free UWB localization. Across 15 single-base scenarios, the protocol restores connectivity in ~ 1 s per scenario; the centroid and weighted variants of the controller are tied on healing latency, while the weighted variant delivers shorter return phases (-8% mean). The comparison under partitioned topologies is left to future work (multi-base and base-failure extensions mentioned in Section VI). Limitations include the need for at least one in-range peer for a lost drone, dependence on anchor density for localization, and a kinematic mobility model that ignores aerodynamics, and multipath.

REFERENCES

- [1] Q. Ouyang, Z. Wu, Y. Cong, and Z. Wang, "Formation control of unmanned aerial vehicle swarms: A comprehensive review," *Asian Journal of Control*, vol. 25, no. 1, pp. 570–593, 2023.
- [2] M. Younis, I. F. Senturk, K. Akkaya, S. Lee, and F. Senel, "Topology management techniques for tolerating node failures in wireless sensor networks: A survey," *Computer Networks*, vol. 58, pp. 254–283, 2014.
- [3] A. A. Abbasi, M. F. Younis, and U. A. Baroudi, "Recovering from a node failure in wireless sensor-actor networks with minimum topology changes," in *2011 IEEE International Conference on Communications (ICC)*. IEEE, 2011, pp. 1–6.
- [4] F. Senel and M. Younis, "Optimized connectivity restoration in a partitioned wireless sensor network," in *2011 IEEE Global Telecommunications Conference (GLOBECOM)*. IEEE, 2011, pp. 1–5.
- [5] R. Olfati-Saber, "Flocking for multi-agent dynamic systems: Algorithms and theory," *IEEE Transactions on Automatic Control*, vol. 51, no. 3, pp. 401–420, 2006.
- [6] O. Khatib, "Real-time obstacle avoidance for manipulators and mobile robots," *The International Journal of Robotics Research*, vol. 5, no. 1, pp. 90–98, 1986.
- [7] K. Piemngam, I. Nilkhamhang, and P. Bunnun, "A virtual spring damper method for formation control of the multi omni-directional robots in cooperative transportation," in *2019 11th International Conference on Information Technology and Electrical Engineering (ICITEE)*. IEEE, October 2019, pp. 1–6.
- [8] Q. Chen, S. M. Veres, Y. Wang, and Y. Meng, "Virtual spring-damper mesh-based formation control for spacecraft swarms in potential fields," *Journal of Guidance, Control, and Dynamics*, vol. 38, no. 3, pp. 539–546, 2015.
- [9] S. Hauert, S. Leven, M. Varga, F. Ruini, A. Cangelosi, J.-C. Zufferey, and D. Floreano, "Reynolds flocking in reality with fixed-wing robots: Communication range vs. maximum turning rate," in *2011 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*. IEEE, 2011, pp. 5015–5020.
- [10] G. Vásárhelyi, C. Virágh, G. Somorjai, T. Nepusz, A. E. Eiben, and T. Vicsek, "Optimized flocking of autonomous drones in confined environments," *Science Robotics*, vol. 3, no. 20, p. eaat3536, 2018.
- [11] M. Coppola, K. N. McGuire, C. De Wagter, and G. C. H. E. de Croon, "On-board communication-based relative localization for collision avoidance in micro air vehicle teams," *Autonomous Robots*, vol. 42, no. 8, pp. 1787–1805, 2018.
- [12] M. Hamer and R. D'Andrea, "Self-calibrating ultra-wideband network supporting multi-robot localization," *IEEE Access*, vol. 6, pp. 22 292–22 304, 2018.
- [13] V. Mai, M. Kamel, M. Krebs, A. Schaffner, D. Meier, L. Paull, and R. Siegwart, "Local positioning system using uwb range measurements for an unmanned blimp," *IEEE Robotics and Automation Letters*, vol. 3, no. 4, pp. 2971–2978, 2018.
- [14] Bitcraze AB, "Crazyflie 2.x nano quadcopter," <https://www.bitcraze.io>, 2026.
- [15] ns-3 Consortium, "ns-3 network simulator," <https://www.nsnam.org>, 2026.